Statistics & Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

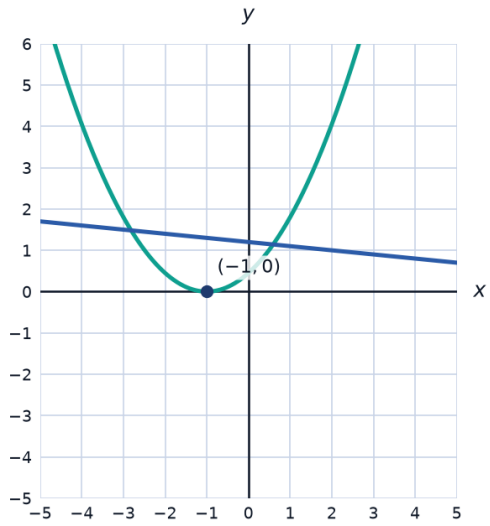
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 4$ in exponential equations, evaluate $f(3)$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Statistics & Probability · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

[AI SL — Specific Topics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Overview · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

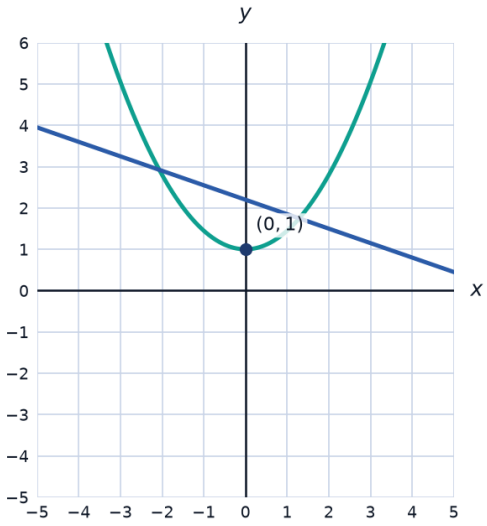
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Number & Algebra · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Calculus · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A bag contains 5 red and 4 green marbles. Three marbles are drawn at random without replacement. Find the probability that: (a) all three are red, (b) at least two are red.

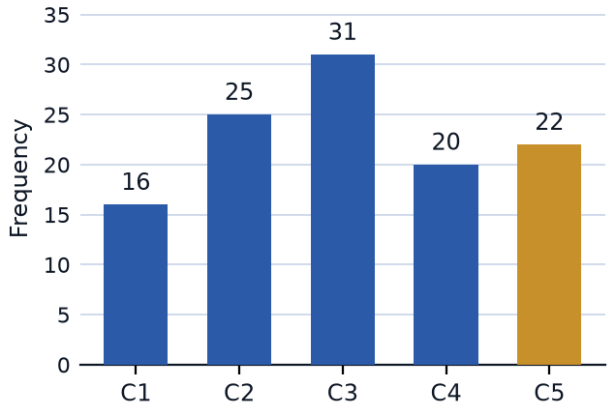
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Events A and B are such that $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cup B) = 0.8$. Find $P(A \cap B)$, calculate $P(A|B)$, and determine whether A and B are independent.

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10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

The table shows the time, t minutes, taken by 50 students to complete a homework task. Intervals: $0 < t \leq 10$ (freq: 8), $10 < t \leq 20$ (freq: 14), $20 < t \leq 30$ (freq: 18), $30 < t \leq 40$ (freq: 10). Calculate an estimate of the mean time taken.



11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A continuous random variable X has probability density function $f(x) = 2e^{-2x}$ for $x \geq 0$ and 0 elsewhere. Find the cumulative distribution function $F(x)$ and determine the exact median of X.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Using the same bag (5 red, 3 blue), find the probability of drawing exactly one red ball.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

[Probability & Statistics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

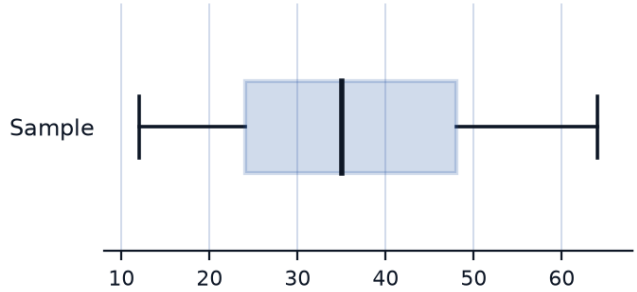
[Statistics & Probability (HL) · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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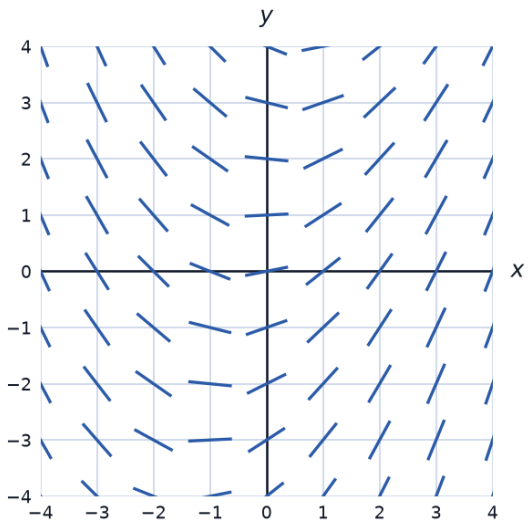
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Geometry & Trigonometry

A continuous random variable X has probability density function $f(x) = kx(2 - x)$ for $0 \le x \le 2$ and $f(x) = 0$ elsewhere. Determine: (a) the constant k; (b) the median of X; (c) the variance $\mathrm{Var}(X)$.



16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Calculus

Rotate the Cartesian axes to eliminate the cross-product term in $5x^2 - 4xy + 8y^2 = 36$. Find the angle of rotation θ, standard form in the rotated system (x', y'), and the lengths of the semi-major and semi-minor axes.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 AI SL — Specific Topics

(IB AA/AI HL) Given the initial value problem $\frac{dy}{dx} = x + y$ with $y(0) = 1$, use Euler's numerical method with step size $h = 0.1$ to approximate $y(0.2)$. Compare your numerical result with the exact analytical solution.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Number & Algebra

(IB AI HL) A market transition matrix $T = (0.7, 0.2; 0.3, 0.8)$ represents weekly customer movement between Brand A and Brand B. Find: (a) the probability that a customer initially with Brand A is with Brand B after 2 weeks; (b) the long-term steady-state market distribution.

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19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Functions

Let $\mathbf{a} = (2; -1; 3)$ and $\mathbf{b} = (1; 4; -2)$. Find the cross product $\mathbf{a} \times \mathbf{b}$, and determine the exact area of the triangle with adjacent sides $\mathbf{a}$ and $\mathbf{b}$.

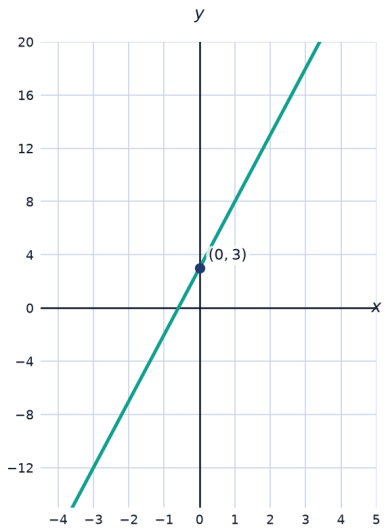
20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.01 Geometry & Trigonometry

Two independent random variables are normally distributed as $X \sim N(50, 16)$ and $Y \sim N(45, 9)$. Find the probability that $X > Y$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 5x + 3$ in logic, evaluate $f(2)$.



a. 13

b. 18

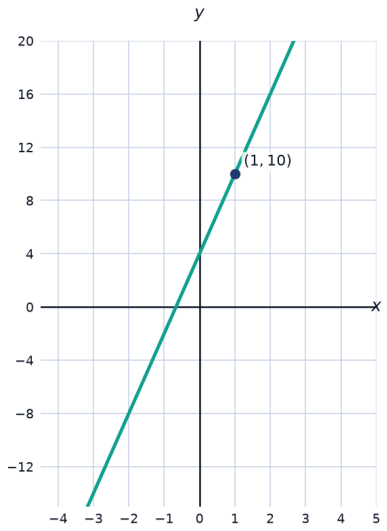
c. 10

d. 17

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 6x + 4$ in proof by deduction, evaluate $f(2)$.



a. 16

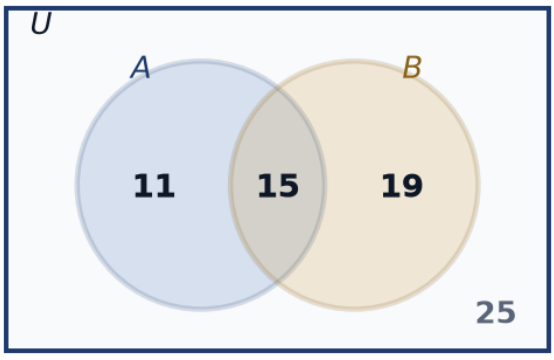
b. 22

c. 12

d. 20

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for logic consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?



a. 15.5

b. 16.5

c. 17.5

d. 19.5

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for proof by deduction consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for exponential equations consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The box plot below shows the distribution of scores for a class of 30 students on a recent test. The interquartile range (IQR) is 22 marks. The median score is 75 marks. The minimum score is 30 marks, and the maximum score is 98 marks. Let the first quartile be denoted by Q_1 and the third quartile be denoted by Q_3. If the upper whisker extends to 98 marks, and the third quartile Q_3 is 86 marks, find the value of Q_1 .

a. 64

b. 75

c. 80

d. 53

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The histogram below shows the distribution of scores for a class of students on a recent quiz. The total number of students who took the quiz is 50. The frequency for the score interval [70, 80) is x. Determine the value of x.

a. 5

b. 10

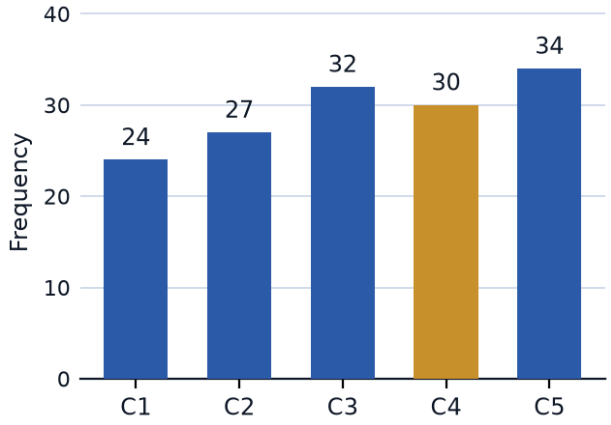
c. 15

d. 20

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28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The bar chart below shows the number of students who chose each of four different sports as their favorite. What is the total number of students surveyed?



a. 50

b. 60

c. 70

d. 80

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The following bar chart shows the number of students who achieved different grades in a mathematics test. The grades are A, B, C, and D. The data is as follows: Grade A: 14 students, Grade B: 22 students, Grade C: 18 students, Grade D: 26 students. What is the total number of students who took the test?

a. 70

b. 80

c. 78

d. 60

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The bar chart below shows the number of students who achieved a certain score on a test. What is the total number of students who scored 6 or 8?

a. 30

b. 28

c. 26

d. 32

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the standard deviation of the data set: $12, 19, 14, 22, 24$, correct to 2 decimal places.

a. 4.98

b. 4.58

c. 4.18

d. 3.78

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The following two-way table shows the results of a survey on favourite sports among a group of students. Calculate the total number of students surveyed.

a. 50

b. 45

c. 60

d. 55

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A bag contains 8 red and 9 blue balls. Two balls are drawn at random without replacement. Find $P(\text{both red})$.

a. $\frac{64}{289}$

b. $\frac{8}{17}$

c. $\frac{7}{34}$

d. $\frac{56}{289}$

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A bag contains 7 red and 8 blue balls. A ball is drawn, replaced, then a second ball is drawn. Find $P(\text{both red})$.

a. $\frac{49}{225}$

b. $\frac{112}{225}$

c. $\frac{8}{15}$

d. $\frac{7}{15}$

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the median of the data set: $8, 30, 32, 34, 48, 53, 54$.

a. 34

b. 33

c. 32

d. 48

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A bag contains 4 blue balls, 8 yellow balls, and 3 green balls. A ball is drawn at random. Find $P(\text{yellow})$.

a. $\frac{8}{15}$

b. $\frac{1}{5}$

c. $\frac{7}{15}$

d. $\frac{4}{15}$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A bag contains 3 red balls, 5 yellow balls, and 8 blue balls. A ball is drawn at random. Find $P(\text{yellow})$.

a. $\frac{11}{16}$

b. $\frac{1}{2}$

c. $\frac{5}{16}$

d. $\frac{3}{16}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Number & Algebra

A bag contains 4 red balls, 3 green balls, and 4 yellow balls. A ball is drawn at random. Find $P(\text{green})$.

a. $\frac{3}{11}$

b. $\frac{2}{11}$

c. $\frac{4}{11}$

d. $\frac{8}{11}$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Functions

A bag contains 3 green balls, 7 red balls, and 4 yellow balls. A ball is drawn at random. Find $P(\text{green})$.

a. $\frac{2}{7}$

b. $\frac{1}{2}$

c. $\frac{11}{14}$

d. $\frac{3}{14}$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.01 Geometry & Trigonometry

A bag contains 4 yellow balls, 6 blue balls, and 3 green balls. A ball is drawn at random. Find $P(\text{green})$.

a. $\frac{6}{13}$

b. $\frac{3}{13}$

c. $\frac{4}{13}$

d. $\frac{10}{13}$

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Calculus

A bag contains 8 yellow balls, 8 red balls, and 4 blue balls. A ball is drawn at random. Find $P(\text{yellow})$.

a. $\frac{9}{20}$

b. $\frac{2}{5}$

c. $\frac{3}{5}$

d. $\frac{1}{5}$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 6.01 AI SL — Specific Topics

A bag contains 6 yellow balls, 2 green balls, and 3 blue balls. A ball is drawn at random. Find $P(\text{green})$.

a. $\frac{9}{11}$

b. $\frac{6}{11}$

c. $\frac{3}{11}$

d. $\frac{2}{11}$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Number & Algebra

A bag contains 3 blue balls, 7 green balls, and 2 red balls. A ball is drawn at random. Find $P(\text{green})$.

a. $\frac{1}{4}$

b. $\frac{7}{12}$

c. $\frac{1}{6}$

d. $\frac{5}{12}$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Functions

A bag contains 6 yellow balls, 4 green balls, and 3 red balls. A ball is drawn at random. Find $P(\text{green})$.

a. $\frac{3}{13}$

b. $\frac{9}{13}$

c. $\frac{4}{13}$

d. $\frac{6}{13}$

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Geometry & Trigonometry

A bag contains 8 green balls, 7 yellow balls, and 6 blue balls. A ball is drawn at random. Find $P(\text{green})$.

a. $\frac{13}{21}$

b. $\frac{2}{7}$

c. $\frac{1}{3}$

d. $\frac{8}{21}$

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